Probabilistic Calibration for MLP Uncertainty

Beyond Deterministic Spray Baselines

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The Core Argument

Goal: Prove that the MLP not only predicts accurate mean penetration trajectories but also provides **statistically reliable dynamic uncertainty bounds (σ)**.

The Limitation of Traditional Baselines:

- Physical baselines (Hiroyasu-Arai, Naber-Siebers) are fundamentally **deterministic**.
- They predict a point estimate, but any provided error boundaries are usually just static, globally computed historical residuals.
- They cannot dynamically adjust confidence based on the instantaneous out-of-distribution (OOD) nature of the specific injection condition.

The Solution: We upgrade our evaluation from simple empirical coverage metrics to formal **Probabilistic Forecasting Diagnostics**.

Evaluation Protocol

Common clean diagnostic set

All models are evaluated on the same $795\,561$ points. For every point i, the diagnostic consumes a scalar observation y_i, predictive mean μ_i, and predictive standard deviation σ_i.

Why this matters

RMSE and MAE only test μ_i. Calibration diagnostics test whether σ_i is statistically meaningful: a useful uncertainty model must be both *calibrated* and *sharp*.

Models compared

Calibrated Hiroyasu–Arai, Naber–Siebers delay, old Stage-3 MLP $\Delta P^{0.25}$, and five new Stage-3 MLP $\Delta P^{0.5}$ seeds.

Source: `MLP/eval/calibration_diagnostics.py`

Metric 1: Reliability Diagram & Expected Calibration Error (ECE)

Step 1: convert each observation to a predictive percentile

$$u_i = F_i(y_i) = \Phi\left(\frac{y_i - \mu_i}{\sigma_i}\right),$$

where F_i is the predicted Gaussian CDF.

Step 2: compare nominal and empirical lower-tail probabilities

$$\hat{p}(\alpha) = \frac{1}{n}\sum_i \mathbf{1}\{u_i \le \alpha\}, \qquad \mathrm{ECE} = \frac{1}{|\mathcal{A}|}\sum_{\alpha\in\mathcal{A}} |\hat{p}(\alpha) - \alpha|.$$

- A calibrated model lies on the identity line $\hat{p}(\alpha) = \alpha$.
- ECE is unitless; lower is better.
- This is stricter than only checking $1\sigma/2\sigma$ coverage because it tests many probability levels.

Metric 2: Continuous Ranked Probability Score (CRPS)

Definition for a Gaussian predictive distribution

$$z_i = \frac{y_i - \mu_i}{\sigma_i}, \qquad \mathrm{CRPS}_i = \sigma_i \left[z_i(2\Phi(z_i) - 1) + 2\phi(z_i) - \frac{1}{\sqrt{\pi}} \right].$$

- CRPS has the same unit as the target: millimetres here.
- It is a proper scoring rule: the expected score is minimized by the true predictive distribution.
- Lower CRPS means a better calibration–sharpness trade-off, not merely a wider interval.
- Sharpness is reported separately as $\mathrm{mean}(\sigma_i)$; sharpness is good only when calibration remains acceptable.

Metric 3: Probability Integral Transform (PIT) & KS Test

PIT values are the same percentiles used for reliability:

$$u_i = \Phi\left(\frac{y_i - \mu_i}{\sigma_i}\right).$$

If the predictive distributions are calibrated, the set $\{u_i\}$ should be uniform on $[0, 1]$.

- U-shaped PIT: too many observations fall in predictive tails, often indicating under-dispersion / overconfidence.
- Hump-shaped PIT: too many observations fall near the predictive median, often indicating over-dispersion / conservative variance.
- A one-sided skew indicates systematic bias in μ.
- With $n = 795\,561$, KS p-values become extremely sensitive; histogram shape and ECE are more informative than treating $p = 0$ as a simple failure label.

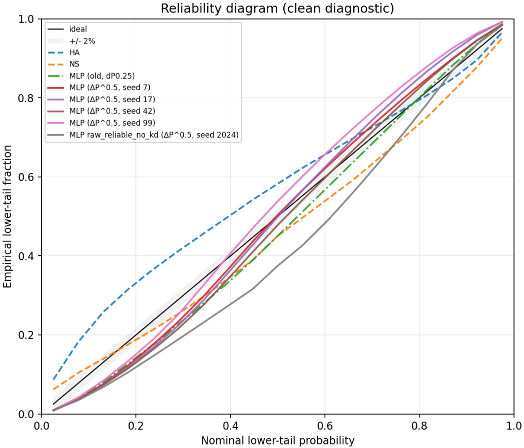
Calibration Results: Summary Table

Model	ECE	CRPS	Sharp.	1σ	2σ
Hiroyasu–Arai calibrated	0.070	5.503	7.478	0.536	0.886
Naber–Siebers delay	0.045	4.894	7.632	0.620	0.893
Stage-3 MLP $\Delta P^{0.25}$, seed 42	0.035	3.409	6.505	0.745	0.979
Stage-3 MLP $\Delta P^{0.5}$, 5-seed mean	0.044	3.245	6.611	0.774	0.981

Readout: the new $\Delta P^{0.5}$ MLP has the best CRPS, so its full predictive distributions score best overall. It is conservative on the clean set (coverage above 0.683/0.954), while the old seed-42 model has the lowest ECE but worse CRPS and no five-seed aggregate.

Source: `MLP/baseline/comparison_reports/calibration_20260509_215557/summary.csv`

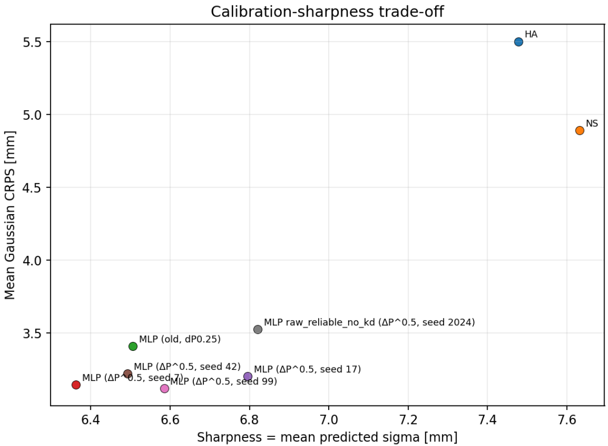
Reliability Diagram: ECE in Picture Form



The diagonal is perfect lower-tail calibration. The $\Delta P^{0.5}$ MLP family stays closer to the diagonal than the deterministic baselines, though the seed-2024 variant visibly drives the five-seed mean ECE upward.

Source: `MLP/baseline/comparison_reports/calibration_20260509_215557/reliability_overlay.png`

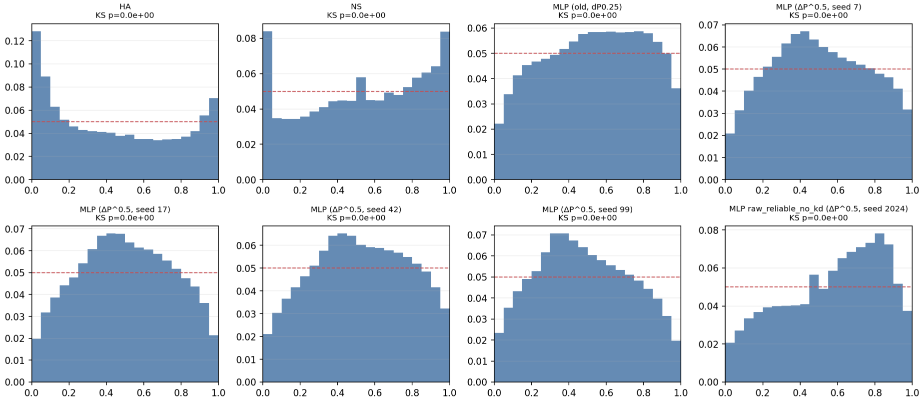
CRPS vs Sharpness



Lower-left is better only if reliability is acceptable. The MLP points dominate HA/NS in CRPS at similar or lower mean σ, showing that the improvement is not produced by simply widening the uncertainty bands.

Source: `MLP/baseline/comparison_reports/calibration_20260509_215557/crps_sharpness_scatter.png`

PIT Histograms



PIT histograms diagnose the failure mode: skew means bias, U-shape means under-dispersion, and centre-heavy histograms mean conservative variance. Because the sample is very large, visual shape is interpreted together with ECE and CRPS.

Source: `MLP/baseline/comparison_reports/calibration_20260509_215557/pit_histograms.png`

Conclusion

Thesis Contribution: By integrating Reliability Diagrams, CRPS, and PIT histograms, we provide a complete suite of diagnostics (as implemented in `calibration_diagnostics.py`).

Takeaway:

Unlike Naber–Siebers or Hiroyasu–Arai models which output rigid curves plus global residual bands, the MLP acts as a heteroscedastic probabilistic surrogate. The current evidence is strongest in CRPS and coverage: the $\Delta P^{0.5}$ five-seed MLP has the best CRPS and conservative clean-set coverage, while GP-style or deterministic baselines remain useful references for nominal calibration.